
Table of Contents

Setup	1
Parameters	1
Initial Conditions	1
SIR	1
Plotting	1
SEIR	2
Discussion	3
SIR ODE	3
SEIR ODE	4

Setup

```
clc; clearvars; close all; format compact
```

Parameters

```
beta = 0.005;  
mu = 0.1;  
gamma = .3;  
time_int = 0:.1:15;  
N = 500;  
  
params = [beta, gamma, mu, N];
```

Initial Conditions

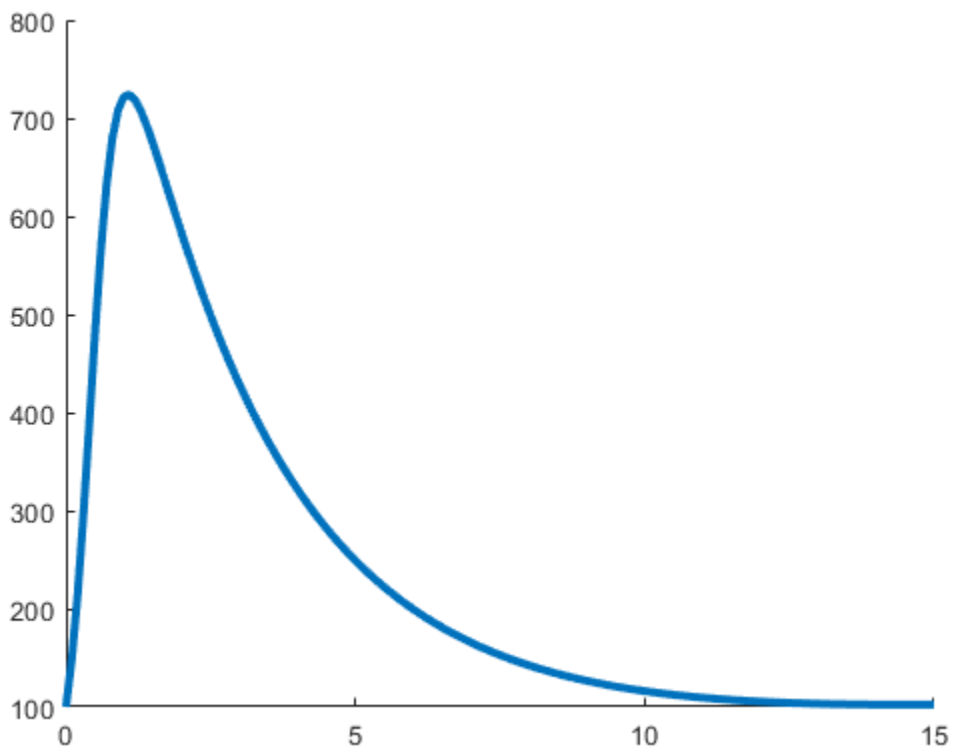
```
So = 900;  
Io = 100;  
Ro = 0;  
Eo = 0;  
alpha = 2.1;  
latency = 1/alpha;
```

SIR

```
init_SIR = [So; Io; Ro];  
  
[T, Y] = ode45(@(T, Y) SIR(T, Y, params), time_int, init_SIR);
```

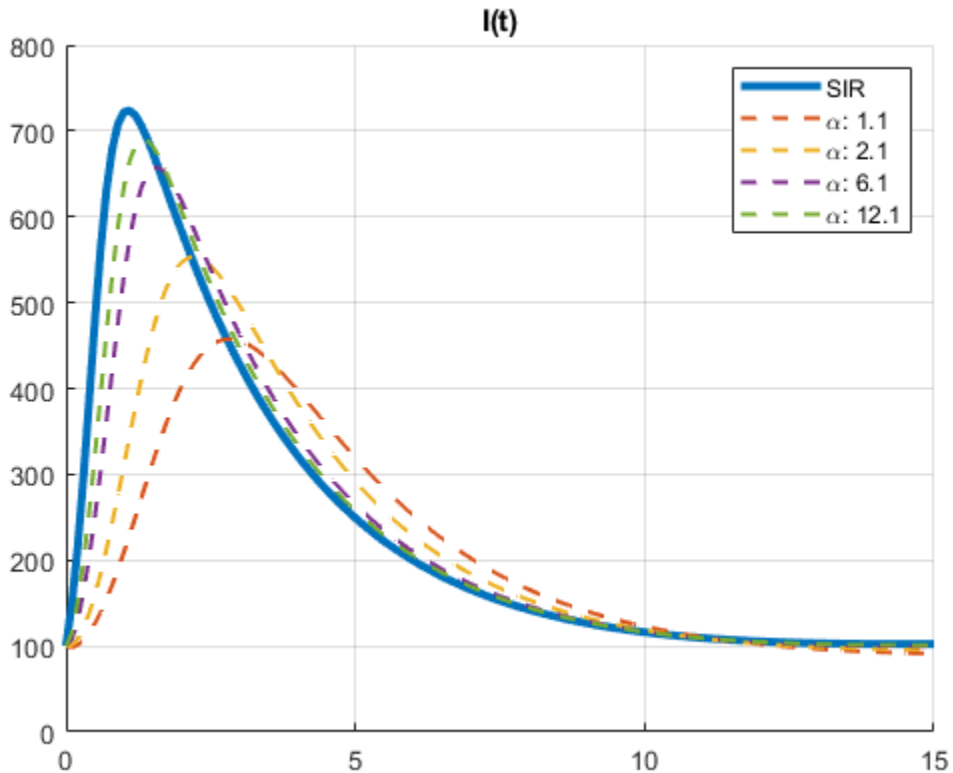
Plotting

```
figure(1);  
hold on  
plot(T, Y(:, 2), 'DisplayName', 'SIR', 'Linewidth', 3);
```



SEIR

```
init_SEIR = [So; Io; Ro; Eo];  
alphas = [1.1 2.1 6.1 12.1];  
  
for i = 1:length(alphas)  
    params(5) = alphas(i);  
    [T, Y] = ode45(@(T, Y) SEIR(T, Y, params), time_int, init_SEIR);  
    plot(T, Y(:, 2), '--', 'DisplayName', '\alpha: ' + string(alphas(i)),  
        'Linewidth', 1.5);  
end  
  
hold off  
grid on  
legend()  
title("I(t)");
```



Discussion

type `qldiscussion.txt`

The figure created demonstrates that with increasing values of alpha, the SEIR model approaches the SIR model. The error between these two curves is logarithmically related to the size of alpha.

The magnitude of E is inversely proportional to the magnitude of alpha. Therefore, with smaller alpha values, the SEIR curve has more latency as the Exposure rate has a larger impact on the behavior. As alpha gets larger and larger, the magnitude of the exposure curve has less and less impact on the infection rate.

SIR ODE

```
function dY = SIR(t, Y, params)
    beta = params(1);
    gamma = params(2);
    mu = params(3);
    N = params(4);

    S = Y(1);
    I = Y(2);
```

```
R = Y(3);

dS = mu*(N-S) - beta*I*S;
dI = beta*I*S - (gamma + mu)*I;
dR = gamma*I - mu*R;

dY = [dS; dI; dR];
```

```
end
```

SEIR ODE

```
function dY = SEIR(t, Y, params)
    beta = params(1);
    gamma = params(2);
    mu = params(3);
    N = params(4);
    alpha = params(5);

    S = Y(1);
    I = Y(2);
    R = Y(3);
    E = Y(4);

    dS = mu*(N-S) - beta*I*S;
    dE = beta*I*S - (alpha + mu)*E;
    dI = alpha*E - (gamma + mu)*I;
    dR = gamma*I - mu*R;

    dY = [dS; dI; dR; dE];
```

```
end
```

Published with MATLAB® R2024b